

A guide to your CSF/PSF data folder

Where to begin, what the files mean, and what still needs your decision.

You do not need to understand all 611 files at once

Start with the README, the two small summary tables and the verification figures. Most of the other files preserve evidence or allow someone else to check the calculations.

What has been prepared

A separate, organized copy of the selected contrast-sensitivity and preferred-spatial-frequency data for manuscript Subjects 1-4. It includes the measurements, derived results, saved resampling distributions and documented code. The original folders remain unchanged.

The package is in `PNAS_Psychophysics_data` within your `Personal_Folder\Git_Folder` directory. The package ZIP contains the same 611 files. These two new PDFs sit alongside the preparation outputs; they do not change the verified package or its checksums.

A short route through the material

If you want to...	Read or open...
Understand the layout	Page 2 of this guide, then README.md.
Understand the calculations	Page 3; follow a real example on page 4.
Look at the data first	Page 5, then validation/figures/.
Understand the unresolved findings	Page 6, then VALIDATION_REPORT.md and AUTHOR_QUESTIONS.md.
Run the checks or prepare a release	Page 7 for code; page 8 for journal readiness.

Three terms to keep in mind

CSF: contrast sensitivity as a function of spatial frequency - how sensitivity changes with the fineness of a pattern.

PSF / f_{pref} : this project uses PSF for preferred spatial frequency, the location of the fitted CSF peak. Here it does not mean point-spread function.

cpd: cycles per degree of visual angle, the unit used for ordinary spatial frequency.

This is a personal reference guide, not a replacement for the deposited data, the manuscript methods or the journal submission form.

1. The folder map

A slash below means "inside this folder". All paths in this guide are relative to the main package folder unless stated otherwise.

Folder	What it contains / why you might open it
data/raw/	Trial records, detected reversals, staircase summaries, display/stimulus settings and preserved source matrices. Start here to trace an observation.
data/source/	Original saved primary CSFs and PSF fit tables, preserved inside gzip with pseudonymous filenames.
data/processed/	Small CSF and PSF summary tables, all saved PSF fits, and separately labelled recomputed curves and coefficients.
data/processed/ bootstrap_csf/	31 compressed tables containing all 310,000 original saved CSF input files, with input and point indices.
data/processed/ figure_source_data/	Numerical tables selected by the supplied Figure 2/3 plotting code. Newly evaluated curves are labelled as recomputed.
data/exclusions/	An explicit log of staircase and fit-point exclusions. Unknown reasons are identified as unknown.
data/calibration_reference/	Four candidate calibration datasets. Their session association and column meanings still need confirmation.
code/ and config/	Relative-path programs, extracted source functions, analysis settings, condition lists and model settings.
validation/	Evidence from the checks: raw-to-CSF comparisons, plots, source hashes, sampled fit checks and diagnostic model replays.

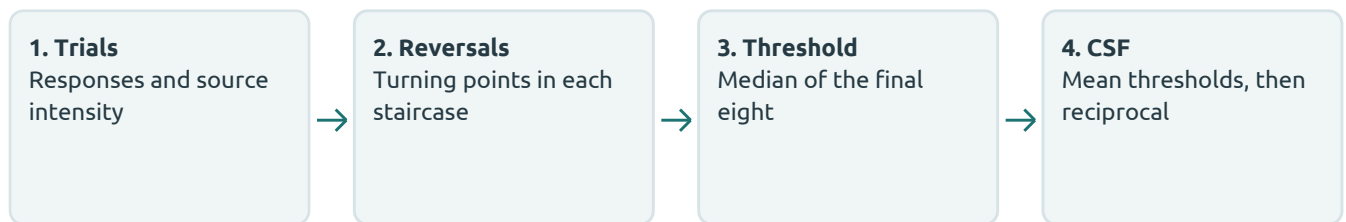
The main documents at the top level

Purpose	Files
Orientation and column meanings	README.md; data_dictionary.csv; metadata_key_dictionary.csv
Findings and unresolved choices	VALIDATION_REPORT.md; AUTHOR_QUESTIONS.md; STAIRCASE_NOTES.md; MODEL_NOTES.md; CALIBRATION_NOTES.md
Traceability and figure coverage	manifest.csv; source_manifest.csv; supplied_materials.csv; source_to_output_mapping.csv; figure_coverage.csv
Running and releasing the package	run_all.md; requirements.txt; environment_provenance.json; CITATION.cff; LICENSE_DATA.txt; LICENSE_CODE.txt; DATA_AVAILABILITY.md; JOURNAL_DATA_CHECKLIST.md

The small .gitignore file tells Git to leave out temporary Python files, local environments and optional new-resampling outputs. It is not a dataset.

2. How the measurements become results

The verified raw-data-to-CSF calculation



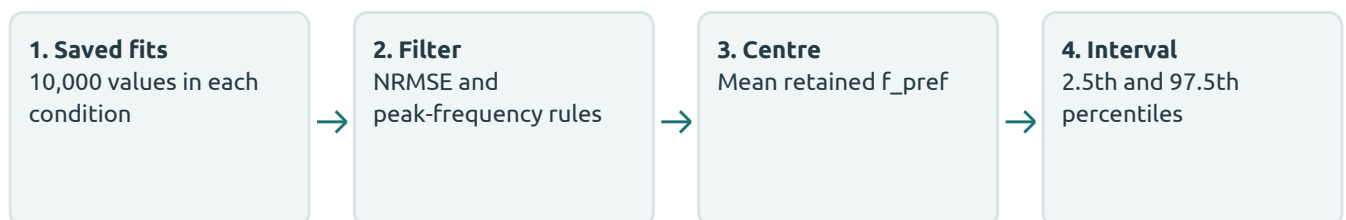
A **staircase** is an adaptive sequence of trials. A **reversal** is a detected change in its direction. The supplied detector identifies these points; it is not assumed to be identical to the experiment's stopping counter.

For each included staircase, take the median of the final eight detected reversal values. At a given spatial frequency, average the thresholds from the included staircases. Contrast sensitivity is **1 divided by this average threshold**. It is not the average of the individual reciprocal thresholds.

A strong verified result

This sequence reproduces all 255 saved primary CSF sensitivities exactly. The staircase plotting script you later supplied independently confirms the median/final-eight rule.

How preferred spatial frequency is summarized



A fitted CSF provides a peak frequency, f_{pref} . The archive contains 10,000 saved fits per condition. The supplied plotting code retains rows with $0 \leq \text{NRMSE} \leq 0.3$ and $f_{\text{pref}} > 1.05 \text{ cpd}$. NRMSE is a normalized fit-error measure.

All 310,000 saved rows remain in the package; 287,067 pass these filters. The reported centre is their arithmetic mean within each condition. The interval is the 2.5th to 97.5th percentile of retained values, using the linear percentile method. These intervals describe reversal-resampling stability; they are not labelled population-bootstrap confidence intervals.

Two distinctions that matter

Preserved versus recomputed: original saved values remain authoritative archive records. Files named `_recomputed` are newly evaluated using the supplied current helper.

Saved input versus saved fit: a row number has not been verified as a link between these two archives. Their mismatch is explained on page 6.

The operational contrast normalization is $(\text{source intensity} - \text{background}) / (1 - \text{background})$. Its physical calibration is still to be confirmed. The spatial-frequency conversion used by the source code is preserved. See `CALIBRATION_NOTES.md`.

3. Follow one real result

Example condition: P01_L010

P01 is manuscript Subject 1. L010 means the 10 cd/m² luminance condition. P02-P04 mean manuscript Subjects 2-4. The private record links these to the original repository participant codes.

Open the small summary tables first

In `data/processed/csf_thresholds.csv`, filter `condition_id` to P01_L010. Each row is one measured spatial frequency in that condition.

Field	Example value	Meaning
<code>spatial_frequency_cpd</code>	0.5	Pattern frequency, in cycles per degree.
<code>contrast_sensitivity</code>	132.245744	Reciprocal of the mean included staircase threshold.
<code>contrast_threshold</code>	0.007561680	Mean included threshold in source-normalized contrast.
<code>included_in_csf_fit</code>	True	This point is used for the baseline CSF fit.

In `data/processed/preferred_frequency.csv`, the same condition has one row summarizing its saved PSF distribution.

Field	Example value	Meaning
<code>f_pref_mean_cpd</code>	3.23016	Mean of retained saved peak frequencies.
<code>percentile_2_5_cpd</code>	2.56550	Lower percentile endpoint.
<code>percentile_97_5_cpd</code>	4.42764	Upper percentile endpoint.
<code>n_saved_fits / n_retained</code>	10,000 / 10,000	All saved fits pass the filters in this example.

Values are rounded here only for readability. The CSV files retain the source precision; the original saved fit files already had four decimal places.

Trace the result back to observations

For the measured CSF point, use `staircase_summary.csv` to find included staircases at the same condition and frequency. Use `staircase_id` to find their reversal rows and then their trial records in `data/raw/`. For the PSF summary, filter `bootstrap_fpref.csv.gz` by the same condition and inspect `retained_for_summary`.

Reading identifiers and missing fields

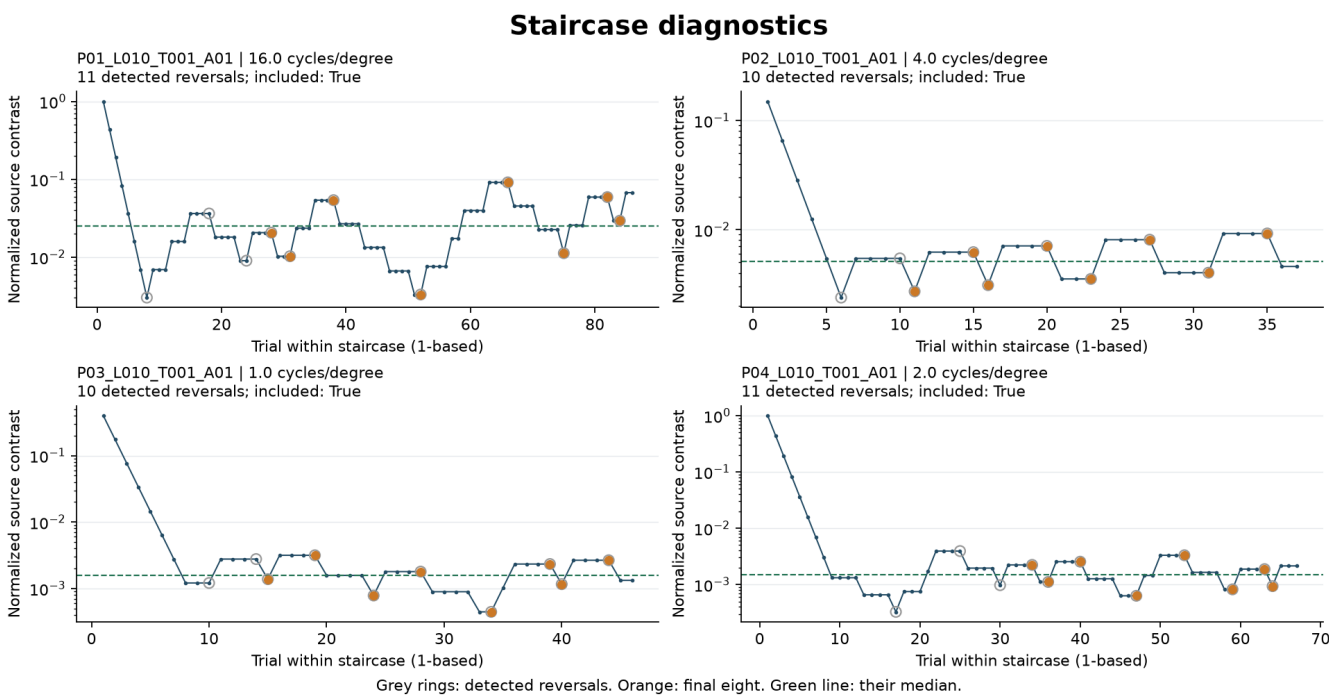
Txxx labels a source trial file; Axx is a staircase within it. These are not acquisition dates. `source_row` and `saved_fit_row` start at 1; `source_input_index` and `source_point_index` start at 0. A blank means unknown/not recorded or not applicable - never zero. Historical `fit_success_recorded` is deliberately blank.

Use `data_dictionary.csv` to check a column by both filename and column name. Some unused original columns are deliberately left as `source_col_XX` because their meanings have not been established.

4. Look at the figures

Open `validation/figures/` for a visual introduction. The PNG files open in ordinary image viewers. The SVG versions can be enlarged while retaining sharp lines and text.

Files	What to look for
<code>csf_p01.png</code> to <code>csf_p04.png</code>	One figure for each subject, showing measurements and current-helper CSF curves across luminance conditions.
<code>preferred_frequency_all_subjects.png</code>	Preferred-frequency summaries across luminance for the four subjects.
<code>staircase_examples.png</code>	Four example staircases showing trial paths, detected reversals and the final-eight threshold calculation.



Existing package figure: `validation/figures/staircase_examples.png`. These are illustrative examples from the archived data, not a new experiment.

Verification plots and manuscript source data

The six figures help inspect the deposited data and calculations. They are not certified final manuscript artwork. The Figure 2/3 numerical source tables are under `data/processed/figure_source_data/`. Their selection follows the supplied plotting code; exact numerical identity with every embedded PDF point or curve has not been certified.

For the current source-to-figure coverage, read `figure_coverage.csv`. Model replay curves are under `validation/model_replay/` and remain diagnostic. Figure 1, Figure 4 and a complete SI computational deposit are not completed by this CSF/PSF package.

5. What the checks do and do not establish

The archive and primary summaries passed their checks. The full documented workflow also ran successfully from a separate copy. Those results do not establish that every historical program setting matches the manuscript.

Finding	What it means for your review
255 exact primary CSF matches	The copied raw trials, reversal detection and median/mean calculation recover the saved primary CSFs.
All saved inputs and fit rows preserved	All 310,000 CSF source checksums and 310,000 PSF row tokens were verified. No missing historical seed or success flag was invented.
Historical bootstrap settings differ	Current scripts select five reversals. Available eight-reversal inputs are consistent with subsets of reversals 2-10 shared across staircases at a frequency. This is evidence about possible values, not proof of the original random algorithm.
Input-to-fit linkage is unresolved	Only 18 of 155 sampled same-position pairs matched all five metrics after four-decimal rounding; 37 matched some saved row. We cannot safely identify saved fit rows as corresponding input iterations.
Reversal counts and exclusions need explanation	Some included staircases have 11 or 12 detected reversals. Two P02/10-cd/m2 staircases are excluded. Recorded exclusions are preserved, but their scientific reasons were not invented.
Model workflow differs from the description	The intrinsic and resonance workflows use different luminance transformations. Intrinsic fits target primary-CSF peaks, and P03 includes auxiliary targets. Separate parameter sets remain separate.
Calibration and acquisition metadata need checking	Some timing metadata say 250 while the manuscript describes 300 ms; units and actual executed settings need confirmation. Candidate calibration files are not yet linked to sessions.

In the model files, f_n means intrinsic frequency and f_r means resonance frequency, both in cpd. The corresponding spatial wavenumber is $k = 2\pi f$. The two model parameter sets come from separate workflows and must not be combined.

How to act on these findings

Start with **AUTHOR_QUESTIONS.md**. Recover or identify the final historical source if possible, and reconcile the Methods text and final figure inputs with the evidence. If a fresh analysis is needed, keep it as a new documented version. Do not silently substitute new outputs for the preserved archive.

Input index 0 is the deterministic primary CSF; each condition has 9,999 additional saved inputs. The manuscript states 10,000 resampling iterations. This discrepancy is also recorded in **VALIDATION_REPORT.md**.

6. Opening files and running the code

File ending	How to use it
.md	A readable text document. Open it in a text editor or a Markdown viewer.
.csv	A table with a header. Open a copy in a spreadsheet or import it into analysis software. Avoid saving over the archive from a spreadsheet, which can change formatting or precision.
.csv.gz / .tsv.gz	A compressed table. Use software that reads gzip, or extract a copy. Preserved source TSV files have no header; column order is documented in the dictionary.
.json / .txt / .cff	Settings, notes or citation metadata stored as text. .cff is citation information for repository tools.
.py / .png / .svg	Python program / ordinary image / scalable vector image.

The programs by purpose

Under code/	Purpose
tests/verify_package.py	Checks hashes, reconstructed source bytes, saved PSF tokens and dictionary coverage. It does not modify the data.
analysis/reproduce.py	Checks raw trials and reversals, reproduces primary CSFs and PSF summaries, and refits current-helper CSF curves.
make_figures/	Plots the data and staircase examples.
bootstrap/audit_saved_fits.py	Compares a fixed sample of saved input files with saved fit rows; reports the historical mismatch.
model/replay_supplied_models.py	Replays the supplied model functions as diagnostic outputs; does not certify final manuscript agreement.
bootstrap/resample_from_reversals.py	Optional new sensitivity analysis. Requires an explicit seed and scheme; it is not the historical generator.
run_all.py	Runs the documented verification, reproduction, plotting, sampled audit and model replay in sequence.

If you want to run it

Work in a fresh copy of the package and follow `run_all.md`. The documented environment uses Python 3.12 and the versions in `requirements.txt`. From that copy's main folder, the complete driver is `python code/run_all.py`. You do not need to run it merely to read the tables or plots.

The manifest is a list of file fingerprints. Run its checks before regeneration: new timings, plots or other validation outputs can legitimately change file hashes. A mismatch after editing means a file changed; it is not by itself a scientific error. Keep the prepared ZIP as an unchanged reference.

A new formal release should receive updated documentation, checksums and a version. The original analysis folders are not needed by these package-relative programs.

7. Is the folder ready for PNAS?

My assessment

The folder is well organized and uses suitable, readable data formats. It is a strong CSF/PSF package for author and reviewer inspection. I would resolve the documented scientific and release issues before describing it as the complete, final public deposit for the paper.

What the journal guidance supports

The official National Academy of Sciences PNAS template lists CSV, text and GZ among dataset formats. It calls for data, materials and associated code/protocols to be available in a public repository upon publication, with access for editors and reviewers if requested during evaluation. It also calls for a sharing statement and dataset citations. These requirements support the format and documentation approach used here. [1]

This is a format/readiness assessment, not advance journal approval. The official PNAS author-center pages returned access errors (HTTP 403) during the 6 September 2026 check; the accessible official NAS template was used instead. Confirm the current portal and full policy before submission. [1-3]

A practical order for finishing the release

Stage	Action
1. Scientific reconciliation	Resolve the bootstrap method/input linkage, reversal/stopping description, exclusions and model findings. Confirm acquisition/calibration details and align the paper with the final analysis.
2. Whole-paper coverage	Identify remaining model, physiological and SI materials needed to support the complete paper. Check figure_coverage.csv.
3. Author release decisions	Confirm public-sharing permission, choose data/code licenses, and finalize author/contact/citation details. The existing license files are explicitly pending.
4. Deposit and cite a fixed version	Select the public repository/archive, verify the files and access, and update the Data Availability statement and dataset citation with a real record and version. A local folder or ZIP alone is not a public deposit.

GitHub can hold the working repository. For the paper, a permanently archived, versioned record such as Zenodo or OSF is a preparation recommendation, not a journal mandate verified here. No repository has been created or uploaded in this task.

Keep private material separate

PRIVATE_provenance_DO_NOT_UPLOAD.zip contains participant linkage and private source details. It is outside the candidate folder and public-candidate ZIP. The PDFs are reading aids; the machine-readable datasets remain the research deposit.

Sources and files for the next step

[1] Official NAS PNAS research-report template (2025), hosted on Overleaf: SI Datasets and Data Availability sections. Accessed 6 September 2026.

[2] PNAS: Submitting Your Manuscript (direct retrieval blocked).

[3] PNAS: Editorial and Journal Policies (direct retrieval blocked).

Next, read AUTHOR_QUESTIONS.md and DATA_AVAILABILITY.md. The proposed availability wording is conditional; the repository is not public.